

Aakash Kumar Agarwal

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PROFILE

AI Research Engineer and MS by Research graduate from IIT Bombay, focused on **LLMs, post-training, machine unlearning, reasoning, multimodal AI, and safety**. First/equal-contribution author of papers at **AAAI 2026** and **EMNLP Main 2025**, with research experience at Fujitsu Research and IIT Bombay.

EDUCATION

Indian Institute of Technology Bombay Mumbai, India
MS by Research, Computer Science (AI/ML); CPI: 8.83 2023 – 2026
– **GATE Computer Science 2023:** All India Rank **156** among 75,680 candidates.

SELECTED PUBLICATIONS

- **Aakash Kumar Agarwal**, Om Chaudhari, Nihar Ranjan Sahoo, Abhinav Joshi, Ashutosh Modi. **Learning to Forget: Hypernetwork-Driven Sharded Unlearning in LLMs**. *ACL ARR 2026, under review*.
- **Aakash Kumar Agarwal***, Moyuru Yamada*. **Hypothesis-Driven Reasoning for Large Language Models**. *AAAI 2026 Main Conference*. *Equal contribution.
- **Aakash Kumar Agarwal***, Saprativa Bhattacharjee*, Mauli Rastogi, Jemima S. Jacob, Biplab Banerjee, Rashmi Gupta, Pushpak Bhattacharyya. **ReDepress: A Cognitive Framework for Detecting Depression Relapse from Social Media**. *EMNLP Main 2025*. *Equal contribution.

EXPERIENCE

Fujitsu Research Bangalore, India
AI Research Engineer Intern May 2025 – Jul 2025

- Developed **Hypothesis-Driven Reasoning (HDR)**, an iterative generate-and-verify framework for improving LLM reasoning over multimodal observations.
- Built a two-stage memory consolidation pipeline with **LangChain/LangGraph** and introduced an **Experience-Transfer** task to evaluate compositional generalization in unseen multimodal settings.
- Evaluated HDR on GPT-4o, Qwen2.5, and Gemma3, achieving near-oracle performance and up to **56.7% improvement** over baselines.

CFILT Lab, IIT Bombay Mumbai, India
Student AI Researcher Jan 2024 – 2026

- Conducted research on **NLP for mental health, LLM safety, multilingual adversarial robustness, and reasoning**, including large-scale LLM-based social-media analysis.

Accenture Gurugram, India
Software Developer May 2019 – Jul 2021

- Developed and maintained production software applications across requirements, implementation, testing, and deployment.

SELECTED PROJECTS

Machine Unlearning for LLMs via Hypernetwork-Generated LoRA IIT Bombay / IIT Kanpur

- Developed **HyperLoRA**, a hypernetwork generating per-shard LoRA adapters from learned shard embeddings, enabling auditable unlearning through $O(1)$ embedding deletion while keeping the base LLM frozen.
- Evaluated five adapter composition/routing strategies on TOFU with LLaMA-3.2; deployable routing achieved **0.80 AGG**, comparable to retraining from scratch (**0.81**) and substantially above the strongest tuned post-hoc baseline (**0.53**).

ReDepress: Depression Relapse Detection from Social Media MS Research, IIT Bombay

- Constructed a longitudinal dataset from **10 years of Reddit data** using multi-stage LLM filtering and clinical expert annotation; engineered cognitive-bias features capturing attention, interpretation, memory, and rumination.
- Built transformer-based temporal models over chronologically ordered posts; integrating cognitive biases improved accuracy from **0.81 to 0.88** and F1 from **0.76 to 0.84**.

TECHNICAL SKILLS

LLMs/ML: PyTorch, Hugging Face, Transformers, Fine-tuning, LoRA/PEFT, DPO/GRPO, RAG, LLM Evaluation, Multimodal LLMs, Machine Unlearning, LLM Safety

Agentic AI: LangChain, LangGraph, AI Agents, MCP, Tool Calling

Engineering: Python, C++, Scikit-learn, NumPy, Pandas, FastAPI, SQL, Git/GitHub, Docker, Linux/Bash